

retiree portfolios, even when the results are based on the excellent performance of stocks found in the US historical record.

This exhibit has several advantages over the corresponding Trinity study shown before, despite being based on the same underlying data. Exhibit 4.8 uses Monte Carlo simulations, which, as I mentioned earlier, count each year of the historical data equally. It is not subject to the bias against bonds in the historical simulations resulting from an overweighting of the middle historical years. The information is more directly useful than portfolio success rates, as the exhibit directly links acceptable failure rates with maximum withdrawal rates and optimal asset allocation.

The results also fit better with real-world investing. With reduced risk capacity and tolerance, stock allocation recommendations for retirees tend to be lower than the 50–75 percent suggested in historical simulation studies. The exhibit shows how a wide range of asset allocations work nearly as well as the optimum. And since the exhibit is based on Monte Carlo simulations, it can be easily customized to allow changing capital market expectations and asset class choices. Indeed, this was my objective for writing the initial research article. I wished to provide the opportunity for readers to see how their own assumptions would impact sustainable spending rates. Users of the article would not be beholden to the assumptions and choices made by the authors of various research articles. The underlying exhibit remains the same, but you are free to develop your own efficient frontiers and map these onto the exhibit to observe the implications for retirement spending.

Further Reading

Bengen, William P. 1994. “Determining Withdrawal Rates Using Historical Data.” *Journal of Financial Planning* 7 (4): 171–180.

Cooley, Philip L., Carl M. Hubbard, and Daniel T. Walz. 1998. “Retirement Savings: Choosing a Withdrawal Rate That Is Sustainable.” *American Association of Individual Investors Journal* 20 (2): 16–21.

Hopewell, Lynn. 1997. “Decision Making Under Conditions of Uncertainty: A Wakeup Call for the Financial Planning Profession.” *Journal of Financial Planning* 10 (5).